

Euclid's Elements — Book II

Proposition 2

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

1 Introduction

Proposition II.2 is the first place where the rectangle calculus established in II.1 is reused rather than developed further. The classical proposition is visually elementary: a square on a straight line is cut by a parallel through an arbitrary point of that line, and the two resulting rectangles fill the square. The formal reconstruction preserves exactly that geometric content, but its proof architecture is deliberately different from Euclid's.

Euclid proves II.2 directly from a square construction and a parallel through the point of division. The current Lean theorem instead imports `Proposition2_1` and applies the representative-independent binary form of II.1. One additional operation is needed: Euclid names each part rectangle with the whole side first, whereas the II.1 cut theorem returns the divided part first. The proof therefore rotates each rectangle representative by one vertex, applies II.1, and then transports the result back to Euclid's original order.

This makes II.2 a particularly clean test of the Book II API. Two algebra-like laws are already available, but neither is primitive algebra:

- additivity is realized by an exact rectangle dissection;
- exchange of the two containing segments is realized by cyclic rotation of a rectangle.

No multiplication of segment lengths and no numerical area function occurs in the theorem.

2 Euclid's Statement and Classical Configuration

2.1 The statement

In Heath's wording, Proposition II.2 states:

If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

Here “both of the segments” means the two rectangles obtained by pairing the whole line with each of the two parts. If AB is cut at C , the geometric statement is therefore

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) = \text{Sq}(AB).$$

The notation is documentary shorthand only. It does not introduce a number-valued rectangle function.

2.2 The classical diagram

Euclid takes a straight line AB cut at C , constructs the square $ADEB$ on AB by I.46, and through C draws CF parallel to the two opposite sides AD and BE by I.31 [?]. The square is thereby partitioned into two rectangles. One is contained by the whole AB and the segment AC ; the other is contained by the whole AB and the segment CB .

The current Lean configuration uses different letters:

classical role	Lean role
AB	BC
C	M
$ADEB$	$BCED$
CF	ML

Thus $B - M - C$, the outer figure $BCED$ is a square, and the cut ML divides it into the left and right rectangular pieces. The formal exact-dissection interface additionally carries the diagonal/cut intersection X ; that genuinely new auxiliary state is shown later, at the II.1 call where it is used.

Diagrammatic audit. Three pictures are retained. Figure 1 records Euclid’s source construction. The proof architecture later shows the two cyclic rectangle-representative changes because those are the distinctive local representation step in Lean, and it separately shows the configured II.1 square cut with the hidden witness X . There is no picture for recognizing the square as a rectangle contained by BC, BC , because that changes only the predicate under which the same square is viewed. Likewise the final addition, transitivity, and symmetry introduce no new geometric configuration.

3 Classical Proof

The classical proof has only four mathematical steps.

First, construct the square on the whole segment. Second, draw through the point of section a line parallel to the appropriate pair of sides of the square. Third, observe that this line partitions the square into two rectangles. Finally, identify those rectangles by Book II Definition 1: their adjacent sides are the whole segment together with the two parts of the divided segment.

In the standard lettering this gives

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) = \text{Sq}(AB).$$

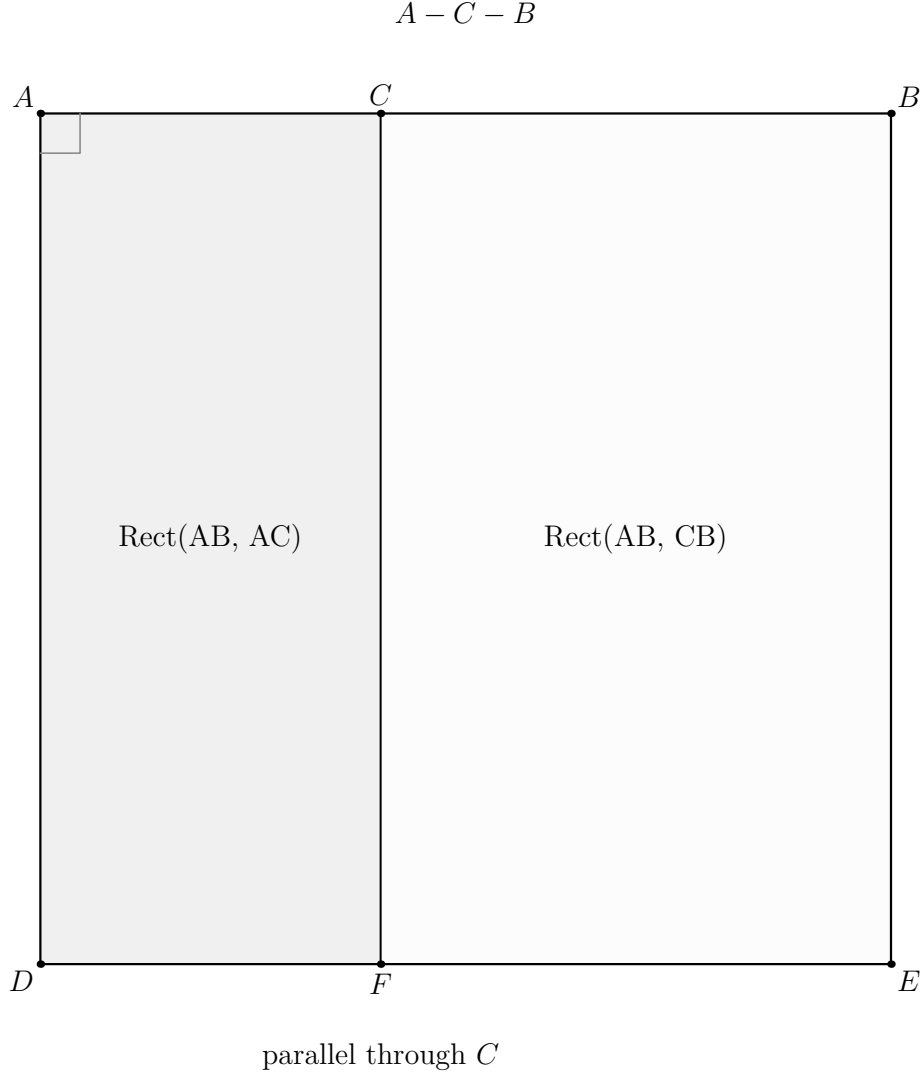


Figure 1: The classical source configuration for II.2. The square on AB is cut at C by the parallel CF . The two visible subfigures are the rectangles contained by AB, AC and AB, CB . No diagonal or intersection witness is part of Euclid's local argument.

Euclid does not invoke II.1 in this proof. The local classical dependency graph is instead

$$\begin{array}{c}
 \text{I.46: construct the square on } AB \\
 \Downarrow \\
 \text{I.31: draw the parallel through } C \\
 \Downarrow \\
 \text{II.Def.1: identify the two rectangles} \\
 \Downarrow \\
 \text{the two pieces constitute the square.}
 \end{array}$$

This fact is important for documentation. The formal theorem proves the same geometric identity, but it is not a line-by-line formalization of Euclid's historical derivation.

4 The Geometric Identity

In the current Lean lettering the square side is BC and the cut point is M . The two Euclidean rectangles are therefore

$$\text{Rect}(BC, BM), \quad \text{Rect}(BC, MC),$$

and the theorem asserts the exact scissors relation

$$\text{Rect}(BC, BM) + \text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Sq}(BC).$$

A modern algebraic shadow is obtained by writing $x = y + z$:

$$x^2 = xy + xz.$$

This is useful as a mnemonic for the pattern, but it is not the object proved by Lean and it is not a claim that Euclid has defined multiplication on segment lengths. The formal terms remain rectangle representatives and their exact scissors relation.

5 Reusable Book II Infrastructure

II.2 imports

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction. Its proof consumes four pieces of the existing Book II infrastructure.

5.1 A square as a rectangle contained by the same side twice

From

```
hSquare : IsSquare Geo B C E D
```

the theorem

```
square_is_rectangle_contained_by_side
```

produces

```
IsRectangleContainedBy Geo B C E D B C B C
```

so the concrete square is recognized synthetically as a rectangle contained by BC, BC .

5.2 The binary form of II.1

The reusable distributive kernel is

```
euclid_proposition_2_1_two
```

whose geometric content is

$$\text{Rect}(BC, PQ) \simeq_{\text{sc}} \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

when $B - M - C$ and the corresponding rectangular cut data are supplied.

For II.2 the fixed segment is specialized to $PQ = BC$. Thus the right-hand pieces naturally arrive in the order

$$\text{Rect}(BM, BC), \quad \text{Rect}(MC, BC).$$

5.3 Exchanging the containing-side order

Euclid’s statement uses the opposite order:

$$\text{Rect}(BC, BM), \quad \text{Rect}(BC, MC).$$

The current proof does not treat this as “commutativity of multiplication.” It calls

```
rectangle_contained_by_swap
```

on each arbitrary representative.

The theorem rotates a rectangle

$$A B C D \mapsto B C D A,$$

returns the corresponding `IsRectangleContainedBy` fact with the two containing segments exchanged, and simultaneously returns a `HilbertScissorsEq` relating the two cyclic presentations.

The higher-level theorem `rectangle_contained_by_comm` belongs to the same rectangle API, but the public II.2 proof uses `rectangle_contained_by_swap` directly because its returned data are exactly the swapped representatives and exact scissors transports needed for the subsequent II.1 call.

6 Proof Architecture of the Reconstruction

The Lean proof has four stages.

6.1 Step 1: recognize the square

The square hypothesis is converted to the “contained by” language:

```
have hSquareContained :
  IsRectangleContainedBy Geo
    B C E D B C B C :=
square_is_rectangle_contained_by_side
  Geo B C E D hSquare
```

The rectangle structure and the side congruence $CE \cong BC$ are then extracted from this fact.

6.2 Step 2: rotate the two arbitrary part representatives

The two hypotheses supplied in Euclid's order are

```
hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M

hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C
```

The current source then applies `rectangle_contained_by_swap` separately to `hPart1` and `hPart2`. Each call returns a swapped `IsRectangleContainedBy` fact and an exact scissors equality for the cyclically rotated representative. The resulting representatives are $V_1V_2V_3V_0$ and $W_1W_2W_3W_0$.

The mathematical result is that the rotated representatives now have the side orders required by the II.1 binary theorem.

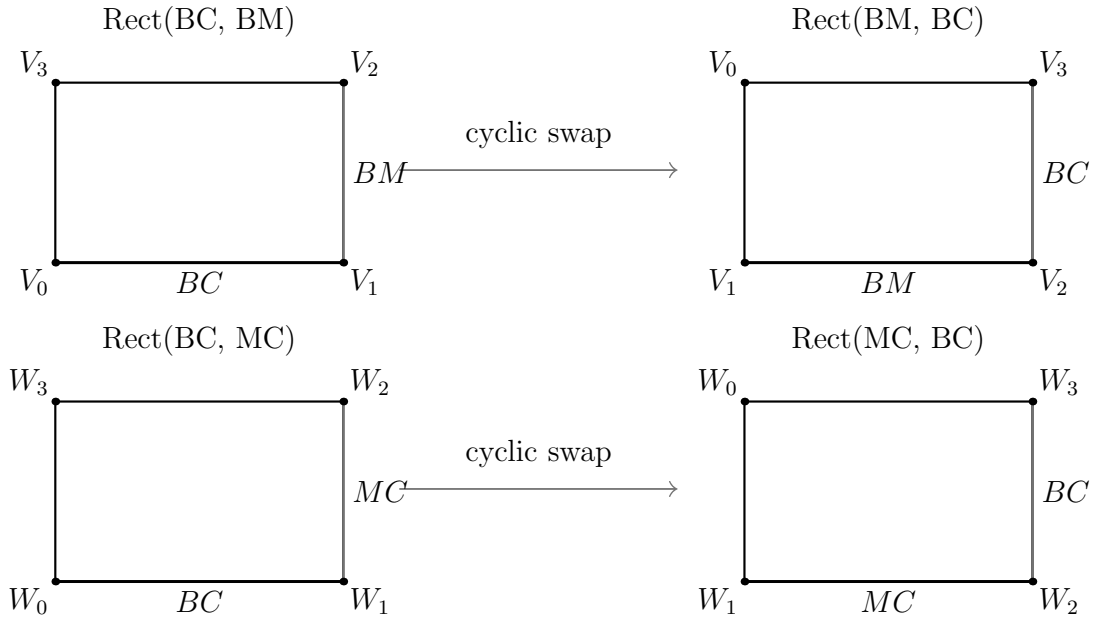


Figure 2: The two independent representative changes performed in Step 2. The first changes the presentation of a rectangle contained by BC, BM to one contained by BM, BC ; the second does the same for BC, MC . The quadrilateral is not replaced by a new metric figure: its vertices are cyclically re-indexed, and `rectangle_contained_by_swap` supplies the exact scissors certificate for that re-presentation.

6.3 Step 3: apply II.1 with fixed segment BC

The central call is

```
have hSplit :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo B C E D)
```

```

      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
        hilbertParallelogramTerm Geo W1 W2 W3 W0) :=
euclid_proposition_2_1_two
  Geo
  B C D E L M X
  B C
  hRect
  hCE_BC
  hBMC
  hDLE
  hDXC
  hMXL
  hLeftPar
  hRightPar
  B C E D
  V1 V2 V3 V0
  W1 W2 W3 W0
  hSquareContained
  hPart1Swap
  hPart2Swap

```

This is the whole mathematical core of the reconstruction:

$$\text{Sq}(BC) \simeq_{\text{sc}} \text{Rect}(BM, BC) + \text{Rect}(MC, BC).$$

6.4 Step 4: return to Euclid's ordering

The two exact rotation equalities are added:

```

have hPartsBack :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
      hilbertParallelogramTerm Geo W1 W2 W3 W0)
    (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
      hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate1)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate2)

```

Transitivity composes `hSplit` with `hPartsBack`, and the final symmetry reverses the result into the order used by the public theorem: sum of the two Euclidean rectangles on the left, square on the right.

Diagrammatic audit. No additional figure is inserted for Step 4. At this point the geometric cut and both representative rotations have already been displayed. The remainder is only exact-scissors addition of the two rotation certificates, transitivity with the II.1 split, and a final symmetry; drawing a new physical arrangement would incorrectly suggest a further dissection that does not occur in the source.

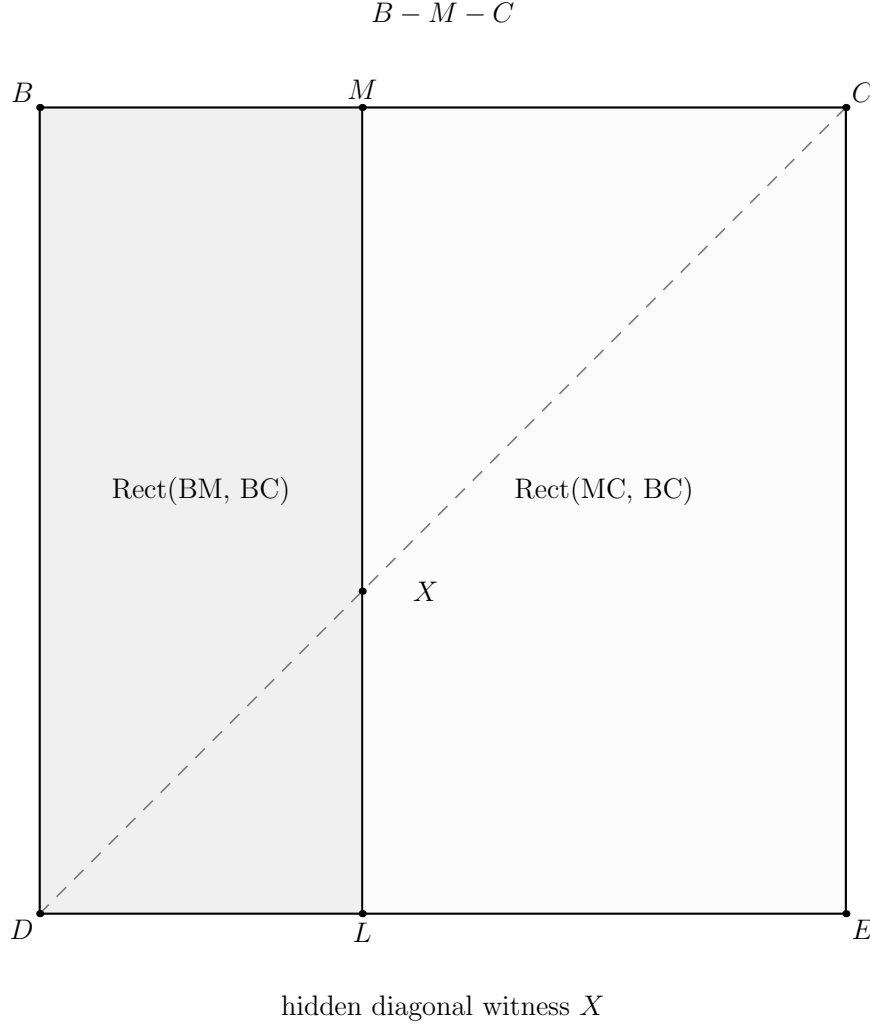


Figure 3: The actual configured square-cut state consumed by the II.1 binary theorem. The order is $B - M - C$, the cut reaches the opposite side at L , and the dashed diagonal DC meets that cut at X . Thus $D - X - C$ and $M - X - L$ are visible here as the hidden formal geometry needed by the inherited exact scissors decomposition. The two pieces are read in the II.1 side order BM, BC and MC, BC .

7 Lean Formalization

The current public theorem is the following instantiated diagram theorem:

```

theorem euclid_proposition_2_2
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (B C D E L M X : Geo.Point)

  (hSquare : IsSquare Geo B C E D)

  (hBMC : Geo.Between B M C)
  (hDLE : Geo.Between D L E)
  (hDXC : Geo.Between D X C)
  (hMXL : Geo.Between M X L)
  (hLeftPar :

```



```

    IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

(V0 V1 V2 V3 : Geo.Point)
(W0 W1 W2 W3 : Geo.Point)

(hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M)

(hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C) :
HilbertScissorsEq Geo
(hilbertParallelogramTerm Geo V0 V1 V2 V3 +
  hilbertParallelogramTerm Geo W0 W1 W2 W3)
(hilbertParallelogramTerm Geo B C E D) := by

```

Three aspects of the signature are mathematically significant.

First, the theorem assumes an instantiated square and cut diagram. It is not an existence theorem constructing the square and the parallel cut from only a segment and a cut point. Euclid performs those constructions in the classical proof; the present public Lean theorem starts after they have been supplied.

Second, the two part rectangles are arbitrary representatives satisfying `IsRectangleContainedBy`. Their exact placement in the plane is not fixed by the statement.

Third, the conclusion is the exact relation `HilbertScissorsEq`. It is not merely a conclusion in the more general `HilbertScissorsEquicomplementable` relation. II.2 remains at the exact-dissection level because its proof is ultimately the binary exact split of II.1 plus exact representation transports.

8 Relationship with Proposition II.1

The contrast between the classical and formal dependency graphs is especially clear here.

Euclid's II.2 is proved directly from the square and the parallel cut. The current Lean file instead begins with

```
import CGJteamLab.Proposition2_1
```

and the public proof calls

```
euclid_proposition_2_1_two
```

once.

Thus II.2 is an explicit formal corollary of the binary II.1 API. No new dissection lemma is proved. The only additional mathematical work is conversion between the two possible orders of the containing segments.

This reuse also explains why the symmetric rectangle interface added at II.1 is not cosmetic. Without the side-order transport, the II.1 theorem would produce perfectly correct rectangles whose formal orientation does not match the language of II.2.

9 Relationship with Book I Infrastructure

The direct import is II.1, not a Book I proposition. Nevertheless the inherited chain underneath II.1 contains the Book I geometry on which the rectangle calculus rests.

The exact binary split is built over the scissors decomposition infrastructure developed around I.47. Rectangle structure uses Euclidean parallelogram theory and synthetic right angles; the square hypothesis uses the square infrastructure established in Book I. Consequently the formal proof of II.2 does not replay I.31 and I.46 as Euclid does. Those classical constructions have been absorbed into the established geometric API and into the hypotheses of the instantiated theorem.

10 Formalization Notes and Hidden Geometric Data

The visible square cut hides several obligations that are explicit parameters of the current theorem:

- $B - M - C$, fixing the orientation of the cut on the square side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the cut with the diagonal used by the inherited exact scissors split;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- two arbitrary representatives of the Euclidean part rectangles;
- two cyclic representative changes, each accompanied by an exact scissors equality.

The point X has no role in the elementary verbal statement. It belongs to the current exact-dissection interface inherited from the triangulated scissors proof. It is therefore documented as hidden formal geometry rather than presented as part of Euclid’s original construction.

There is also an important negative observation. No `SameRay`, `SameSide`, or `OppositeSide` obligation appears in `Proposition2_2.lean` itself. The corresponding low-level geometry has already been discharged inside the reusable rectangle and II.1 theorems.

11 Geometry, Content, Representation, and Later Arithmetic

11.1 Geometry

The geometric layer consists of a square, betweenness on two opposite sides, the diagonal/cut witness, and two parallelogram pieces. Congruence of the square side with itself identifies the square as a rectangle contained by BC, BC .

11.2 Content / scissors

The conclusion and every transport used in the proof are exact `HilbertScissorsEq` relations. The proof uses symmetry, transitivity, and addition of exact scissors equalities. It does not weaken to the more general equicomplementability relation.

11.3 Representation

Representation is the distinctive issue in II.2. The theorem accepts arbitrary rectangles for

$$\text{Rect}(BC, BM) \quad \text{and} \quad \text{Rect}(BC, MC).$$

The binary II.1 theorem wants the factors in the reverse order. Cyclic rotation of each representative supplies the required conversion while preserving exact scissors content.

11.4 Later arithmetic / numerical area

Neither layer is used. In particular, the proof does not define a segment product, does not invoke Hilbert’s later segment field, and does not assign real areas to the square or rectangles.

12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore works in the same Euclidean project layer as II.1. No new proposition-specific geometric axiom or content axiom is declared in `Proposition2_2.lean`. Inspection of the current 162-line source shows one theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as “axiom-free.” The theorem inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output was supplied with the source used for this documentation pass, so the transitive axiom list is not reproduced here.

The implementation checkpoint records that the II.2 Lean build had already been checked during theorem development. This documentation workspace does not contain the complete Lean project, so the Lean build is not re-run here; the LaTeX and Asymptote checks are reported separately.

13 Hartshorne’s Semantic Control

Hartshorne places the end of Book I and all of Book II inside Euclid’s theory of area, interpreted synthetically as equal content rather than as an unexplained numerical area function [?]. He also describes Book II as part of an “algebra of areas”: rectangles can play a product-like role because Euclid has no operation that multiplies two line segments to produce another segment.

II.2 fits that interpretation exactly, provided the levels are kept separate. The square on the whole and the two rectangles are geometric figures; their equality is a content statement; the modern identity $x^2 = xy + xz$ is only an interpretive shadow.

The current Lean theorem is locally stronger than the weakest equal-content reading because its configured proof stays at exact scissors equality. That strength comes from the explicit binary dissection inherited from II.1, not from a general principle identifying every equal-content pair with an exact dissection.

14 Hilbert's Layering

Hilbert supplies a second semantic guardrail. His theory separates synthetic plane content from the later arithmetic of segments. Segment multiplication is introduced only in a later arithmetic construction, with a fixed unit segment and parallel geometry; it is not primitive rectangle notation.

The II.2 proof remains entirely on the earlier side of this boundary:

square and rectangle geometry
 $\Downarrow$
 exact scissors dissection
 $\Downarrow$
 representative rotation and transport.

The apparent commutativity required by the formula $xy = yx$ is realized geometrically by `rectangle_contained_by_swap`, not imported from Hilbert's later segment arithmetic.

15 Dependency Summary

The actual Lean dependency chain is

Hilbert incidence + Euclidean plane layer
 $\Downarrow$
 Book I right-angle, square, parallelogram, and scissors infrastructure
 $\Downarrow$
`HilbertRectangle : IsRectangleContainedBy, rectangle_contained_by_swap`
 $\Downarrow$
 Proposition II.1 binary API
 $\Downarrow$
`euclid_proposition_2_1_two`
 $\Downarrow$
 two representative rotations + exact scissors addition/transitivity
 $\Downarrow$
`euclid_proposition_2_2.`

The classical local chain is different:

I.46 $\longrightarrow$ I.31 $\longrightarrow$ square cut into two rectangles $\longrightarrow$ II.2.

New geometric axiom in II.2:	no
New content axiom in II.2:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.2:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Direct side-order transport used:	yes, <code>rectangle_contained_by_swap</code>
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local <code>sorry/admit</code> :	no

16 Conclusion

Proposition II.2 is mathematically a square cut into two rectangles, but formally it does more than repeat the picture. It demonstrates that the II.1 rectangle calculus is already reusable: the binary dissection theorem supplies additivity, and cyclic rectangle rotation reconciles the two possible orders of the containing sides.

This is also the first sharp divergence between Euclid’s local proof graph and the formal Book II graph. Euclid proves II.2 directly from I.46 and I.31. Lean uses II.1 as a library theorem and turns II.2 into a short transport-and-reuse argument. The theorem is therefore not a literal transcription of the classical proof, but it remains entirely synthetic and geometric.

The result can be remembered by the modern shadow

$$x^2 = xy + xz, \quad x = y + z,$$

but the formal content is more precise: arbitrary representatives of the two rectangles contained by the whole side and the two parts are exactly scissors-equal, in sum, to the concrete square on the whole side. No numerical area and no multiplication of segment lengths is required.